

Pacing Guides

Choose the schedule that fits your teaching or learning rhythm. These pacing plans expand on the 3 pacing options introduced in the How to Use page — pick the one that fits and adapt as needed.

1-Week Intensive Plan

For summer camps, spring break, or focused CS immersion. 2–3 activities per day for 5–7 days.

DAY 1 — Foundations of Algorithms

- Activity 1: What is an Algorithm?
- Activity 2: Sequencing Puzzle
- **Discussion:** What's the difference between random steps and an algorithm?

DAY 3 — Conditionals

- Activity 5: Yes/No Conditionals
- Activity 6: Nested Conditionals
- **Discussion:** Draw a decision tree for your morning routine

DAY 5 — Sorting Part 2 & Patterns

- Activity 9: Selection Sort
- Activity 10: Pattern Recognition
- **Reflection:** How is selection sort different from bubble sort?

DAY 7 — Capstone

- Activity 12: Design Your Own Algorithm
- Complete the Vocabulary Review, Mini-Quiz, and Reflection Page
- **Celebration:** Celebration! Kids get to teach someone one concept

DAY 2 — Loops

- Activity 3: Spot the Loop
- Activity 4: Write Your Own Loop
- **Hands On:** Do the Loop Bonus outside (pattern hunt)

DAY 4 — Debugging & Sorting Part 1

- Activity 7: Debug the Robot
- Activity 8: Bubble Sort
- **Hands On:** Do the "Sort Real Things" bonus with household objects

DAY 6 — Integration

- Activity 11: Treasure Hunt Grid
- **Note:** Take a break — this one is meaty!

4-Week Unit Plan

For homeschool, after-school programs, or supplementary CS. 3 activities per week for 4 weeks.

WEEK 1 — Foundations of Algorithms

- Activity 1: What is an Algorithm?
- Activity 2: Sequencing Puzzle
- Activity 3: Spot the Loop
- **Reflection:** Weekly reflection: Where did you see loops today?

WEEK 3 — Debugging & Sorting

- Activity 7: Debug the Robot
- Activity 8: Bubble Sort
- Activity 9: Selection Sort
- **Hands On:** Do the "Algorithm Race" bonus with a family member

WEEK 2 — Loops & Conditionals

- Activity 4: Write Your Own Loop
- Activity 5: Yes/No Conditionals
- Activity 6: Nested Conditionals
- **Discussion:** Design a decision tree for a weekly household task

WEEK 4 — Integration & Capstone

- Activity 10: Pattern Recognition
- Activity 11: Treasure Hunt Grid
- Activity 12: Design Your Own Algorithm
- Complete Vocabulary Review, Mini-Quiz, and Reflection Page

12-Week Semester Plan

For elementary CS class or gentle weekly rhythm. 1 activity per week for 12 weeks.

Week	Activity	Focus
1	Activity 1 — What is an Algorithm?	Definitions
2	Activity 2 — Sequencing Puzzle	Order matters
3	Activity 3 — Spot the Loop	Recognizing repetition
4	Activity 4 — Write Your Own Loop	Writing loops
5	Activity 5 — Yes/No Conditionals	Decision-making
6	Activity 6 — Nested Conditionals	Decision trees
7	Activity 7 — Debug the Robot	Finding bugs
8	Activity 8 — Bubble Sort	First named algorithm
9	Activity 9 — Selection Sort	Comparing algorithms
10	Activity 10 — Pattern Recognition	Finding rules
11	Activity 11 — Treasure Hunt Grid	Integration
12	Activity 12 — Design Your Own Algorithm	Synthesis

Reminder: Every activity has both a Basic (ages 8-10) and Challenge (ages 10-12) version. Kids can revisit any activity in Challenge mode after completing Basic — that's the built-in extension.